Problem: Supply-chain fragility in high-complexity product industries

Importance & relevance

Businesses producing highly complex products (e.g. semiconductors, aircrafts, electric vehicles) rely on global supply chains involving hundreds of specialized firms. These form small-world-like networks, where local supplier clusters connect via a few global integrators. While this structure enables fast coordination, it also creates vulnerability: disruption of key bridging nodes can cripple entire product systems. Understanding how small-world structure affects systemic risk and capability upgrading is vital for firms aiming to stabilize operations while continuing to move into more complex product architectures.

What we already know

There are multiple researches related to such topics. Watts and Strogatz highlighted that Network theory shows that small-world structures reduce transaction costs but increase systemic interdependence (Watts & Strogatz, 1998). Research on supply-chain resilience (Kim et al., 2015, *Supply Chain Management Review*) highlights the importance of redundancy and modularity. Economic complexity studies indicate that as product sophistication rises, interdependencies increase non-linearly (Hausmann et al., 2014). However, there is limited firm-level evidence connecting network metrics (clustering, degree, betweenness) to business resilience and innovation potential in complex product manufacturing.

Niche + research question

The niche is the lack of integrated measures linking small-world supply-chain topology and firm outcomes (innovation performance, risk exposure). Research question: *How does the small-world structure of a firm's supply-chain network affect its capacity to sustain production and innovate under shocks in high-complexity industries?*

Stakeholders & benefits

Apparently anyone who is part of a supply chain of a highly comlex product can be considered a stakeholder of this problem. All the big companies are continuously monitoring the market, the competitiors and the R&D results that can have some influence on their industry. It is a crucial issue for them to be prepared for a shock, a break of the supply chain, so understanding the structure can help them surviving such events. Therefore, it would be benefical for firms to have the possibility to use network diagnostics to identify structural vulnerabilities and design supply chains with optimal connectivity — maintaining efficiency while reducing systemic risk and improving innovation capability.

References

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